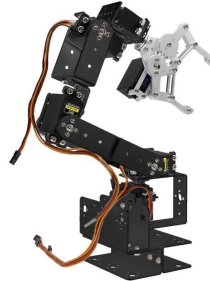


Intro to Robotics: Robot Demo 2 “Forward Kinematics: build a tower”

Name 1:

Name 2:



Demo 2 is worth 100 points. Submit a YouTube video link of your robot. Your name(s) must be permanently written on the robot, and this must be visible in the video, or the demo will not be graded.

This demo will use a servo driver to control at least four motors.

1. **Tape** a large sheet of paper onto your workspace. **Tape** or attach your robot to the workspace. Place three objects to move on the workspace. (*The objects can be small wooden blocks.*) Outline the objects. These are the ‘pick-up locations’.
2. Using the Arduino, hardcode a movement to grasp the object at pick-up location.
3. Using the Arduino, hardcode a movement to move the robot to a “build location”. The drop-off location must be at least 5 cm distance from the pick-up location.
4. Release the object.
5. Stack object 2 on object 1
6. Stack object 3 on object 2

You can get more points by having the Arduino control more degrees of freedom.

The video

- (20 pts) must show your faces & be less than 120 seconds long (you can speed up playback speed). Robot MUST be permanently marked with unique markings.
- (20 pts) Block 1 moved to build location.
- (20 pts) Block 2 stacked on Block 1.
- (20 pts) Block 3 stacked on Block 2.
- If Arduino controls (gripper is 1 DOF)

1 DOF	5 pts total
2 DOF	10 pts total
3 DOF	15 pts total
4 DOF	20 pts total
5 DOF	25 pts total
6 DOF	30 pts total

Enjoy! -Dr. B